

1.3

Applications

Complex Potential Function:

Let $w = f(z) = \phi(x, y) + i\psi(x, y)$ be an analytic function. Then it is called complex potential function. Its real part $\phi(x, y)$ is called velocity potential function ^(potential function) and imaginary part $\psi(x, y)$ is called stream function (flux function).

Problems:

1. If the potential function is $\log \sqrt{x^2 + y^2}$, find the flux function and the complex potential function.

Sol: Let $w = f(z) = \phi(x, y) + i\psi(x, y)$ be the complex potential function.

Then $\phi(x, y)$ is the potential function and $\psi(x, y)$ is the flux function.

Given $\phi(x, y) = \log(\sqrt{x^2 + y^2})$

i.e., $\phi(x, y) = \frac{1}{2} \log(x^2 + y^2)$

Now, $f'(z) = \frac{\partial \phi}{\partial x} + i \frac{\partial \psi}{\partial x}$

$$= \frac{\partial \phi}{\partial x} - i \frac{\partial \phi}{\partial y} \quad (\text{by C-R equations})$$

$$= \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}$$

By Milne-Thomson's method, $f'(z)$ is expressed in terms of z by replacing x by z and y by 0.

Therefore, $f'(z) = \frac{z}{z^2} = \frac{1}{z}$

Integrating, we get

$$f(z) = \log z + C, \quad \text{where } C \text{ is a complex constant}$$

Hence, the required complex potential function is $w = f(z) = \log z + C$

Now, $w = \log z + c$

$$= \log(x+iy) + ik \quad (\text{where } c = ik)$$

Let $x = r \cos \theta$ and $y = r \sin \theta$.

Then, $w = \log(r \cos \theta + i r \sin \theta) + ik$

$$\Rightarrow \phi + i\psi = \log[r(\cos \theta + i \sin \theta)] + ik$$

$$= \log r + \log e^{i\theta} + ik$$

$$= \log(\sqrt{x^2 + y^2}) + i\theta + ik$$

$$= \log(\sqrt{x^2 + y^2}) + i \tan^{-1}\left(\frac{y}{x}\right) + ik$$

Therefore, the required flux function

is $\psi(x, y) = \tan^{-1}\left(\frac{y}{x}\right) + k$.

2. If $w = \phi + i\psi$ represents the complex potential function for an electric field

and $\psi(x, y) = x^2 - y^2 + \frac{x}{x^2 + y^2}$, find the

function $\phi(x, y)$.

Answer: $\phi(x, y) = -2xy + \frac{y}{x^2 + y^2} + c$

(3) In a two-dimensional flow of a fluid, the velocity potential is $\phi(x, y) = x^2 - y^2$. Find the stream function ψ .

Sol: Since $\phi(x, y) = x^2 - y^2$ is velocity potential, it must satisfy the Laplace equation.

$$\text{here } \frac{\partial \phi}{\partial x} = 2x, \quad \frac{\partial \phi}{\partial y} = -2y$$

$$\frac{\partial^2 \phi}{\partial x^2} = 2 \quad \text{and} \quad \frac{\partial^2 \phi}{\partial y^2} = -2$$

$$\text{clearly, } \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

$\therefore \phi$ satisfies the Laplace equation.

Let $w = f(z) = \phi(x, y) + i\psi(x, y)$ be a complex potential function, where $\psi(x, y)$ is the stream function.

Then $f'(z) = \frac{\partial \phi}{\partial u} + i \frac{\partial \psi}{\partial u}$

$$= \frac{\partial \phi}{\partial x} - i \frac{\partial \psi}{\partial y} \quad (\text{by C-R equations})$$

$$= 2x + i2y$$

By Milne-Thomson's method, replacing

x by z and y by 0 .

$$\therefore f'(z) = 2z$$

Integrating, we get

$$w = f(z) = z^2 + C, \text{ where } C \text{ is a Complex Constant.}$$

$$\Rightarrow \phi + i\psi = (x+iy)^2 + C$$

$$= (x^2 - y^2) + i2xy + iC_1$$

$$(C = iC_1)$$

$$= (x^2 - y^2) + i(2xy + C_1)$$

Hence the required stream function

$$\text{is } \psi(x, y) = 2xy + C_1$$

④ If $w = \phi + i\psi$ represents the complex potential for an electric field and $\psi = 3x^3y - y^3$, find ϕ .

Answer: $\phi(x, y) = x^3 - 3xy^2$

⑤ In a two-dimensional flow of a fluid, the stream function is $\psi = \frac{-y}{x^2 + y^2}$. Find the velocity potential (or, potential function).

Answer: $\phi(x, y) = \frac{x}{x^2 + y^2}$.